(H&P). The aim of this study is to determine the accuracy of these initial clinical assessments and the intuitive prognoses that they may inform

Materials/Methods: We evaluated a retrospective, multi-institutional database of 1523 brain metastases in 507 patients who were treated with radiosurgery from 2001-2014. Relevant historical information included comorbidities, performance status (ECOG), and seizure history. Neurologic exam findings included a sensory exam, motor exam, and cognitive function. Primary tumor histology and systemic disease status were excluded from the analysis in order to reflect a clinical situation consistent with evaluating a new patient with intracranial metastases from an unknown primary. Univariable and multivariable logistic regression and Cox regression analyses were used to identify predictors of OS.

Results: A logistic regression model of comorbidities from patient's past medical histories identified cardiac and neurologic comorbidities as being significant predictors of decreased OS ($p=0.01$ and $p=0.029$ respectively). In a multivariable Cox regression model drawn from HPI, OS was predicted by the presence of seizures (OR 1.63, 95%C.I. 1.02-2.60; $p=0.043$) and increased ECOG score (OR 4.76, 95%C.I. 2.4-9.4; $p<0.001$). 21.9% of the patients had neurologic deficits on initial presentation, 9.7% had cognitive decline, and 8.5% had seizures. When neurologic exam findings were included in a multivariable Cox model of all H&P factors, increased ECOG (OR 3.5, 95%C.I. 1.6-7.9; $p=0.001$), cardiac comorbidities (OR 1.77, 95%C.I. 1.2-2.6; $p=0.005$), neurologic comorbidities (OR 1.79, 95%C.I. 1.0-3.1; $p=0.042$), and the presence of seizures (OR 1.8, 95%C.I. 1.05-3.15) predicted worse OS (Figure 1). Neurologic exam findings were not associated with OS. Interestingly, patients with neurologic motor deficits were more likely to have undergone surgical resection ($p<0.001$).

Conclusion: Neurologic deficits are not associated with initial overall prognosis for patients with brain metastases undergoing radiosurgery. Despite the appeal for assuming that patient's with a worse neurologic exam will fare more poorly, our findings confirm that when encountering new patients with brain metastases of unknown primary histology or systemic disease status, their prognosis should be informed by initial historical findings that delineate performance status, past medical history, and seizures.

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2193

Multi-institutional Nomogram Predicting Survival Free From Salvage Whole-Brain Radiation After Radiosurgery for Brain Metastases

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Purpose/Objective(s): Use of radiosurgery (SRS) for primary management of brain metastases is typically motivated by the desire to control intracranial disease and avoid whole brain radiation therapy (WBRT). This study aimed to devise a useful and convenient nomogram that predicts probability of surviving without salvage WBRT after SRS.

Materials/Methods: Multi-institutional data were retrospectively collected from 1045 patients treated with SRS alone for newly diagnosed brain metastases. SRS was generally offered for limited brain metastases (1-4 lesions) in patients with KPS $\geq$70 and life expectancy $\geq$3 mo. Salvage WBRT was typically recommended for intracranial failures with >4 new lesions, KPS <70, or life expectancy <3 mo. WBRT-free survival was defined from date of SRS to date of salvage WBRT or death. Multiple patient, treatment, and disease variables were evaluated. Extracranial disease burden was defined as absent, oligometastatic ($\leq$ 5 lesions), or widespread (>5 lesions). Kaplan-Meier method was used to estimate WBRT-free survival. Cox proportional hazard regression model was used for multivariate analysis. Significant prognostic factors for WBRT-free survival were identified and integrated to build a nomogram that was subjected to bootstrap internal validation.

Results: Median WBRT-free survival among all patients was 8 mo (range: 0.1-139 mo). The following were significant independent predictors for reduced WBRT-free survival ($p<0.05$): older age (HR=1.1 with each decade), more brain metastases (HR=1.1 with each additional lesion), presence of symptoms (HR=1.26), progressive systemic disease (HR=1.34), increasing extracranial disease burden [oligometastatic (HR=1.33), widespread (HR=1.59), each relative to none], and histology [colorectal primary (HR=1.41) and HER2(-) breast primary (HR=1.6), each relative to other histologies]. Trends for HER2(+) breast primary predicting for favorable WBRT-free survival (HR 0.82) and melanoma for shortened WBRT-free survival (HR=1.1) were noted. The nomogram developed estimates probabilities of 6- and 12-mo WBRT-free survival based on coefficients from the aforementioned regression model with Harrell's C concordance statistic of 0.62. The table below displays nomogram output for 3 sample cases.

Conclusion: Appropriate patient selection for upfront SRS alone has important quality of life and cost implications. Our nomogram can be used to optimize management decisions by providing individual prediction of a novel and clinically useful endpoint, WBRT-free survival.

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Poster Viewing Abstracts 2193; Table 1

Case	Age	Histology	# of brain metastases	CNS symptoms	Extracranial disease status	Extracranial disease burden	6-mo probability of WBRT-free survival	12-mo probability of WBRT-free survival
1	50	HER2(+) breast	1	Absent	Stable	None	>80%	>70%
2	60	NSCLC	3	Present	Stable	Oligometastatic	55%	35%
3	70	Colorectal	5	Present	Progressive	Widespread	<20%	<5%

2195

^{106}Ruthenium Eye Plaque Brachytherapy in the Management of Medium Sized Uveal Melanoma

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Purpose/Objective(s): We evaluated patients with uveal melanoma in our ophthalmic university clinic in the past years. The results revealed an incidence of 11 per million in the population in north Germany, which is higher than the incidence of the white population in the U.S. (5.1 per million). Nineteen percent of the evaluated patients, whose visual acuity could no longer be saved, underwent an enucleation. Depending on the size and location of the tumor, 41% of the patients received proton therapy while 37% received Ru-brachytherapy. This is a report of our experience of Ru-brachytherapy.

Materials/Methods: From 2007 to 2014, 35 Patients with medium sized-uveal melanoma were treated with ^{106}Ruthenium eye plaque brachytherapy. At the time of diagnosis the median tumor height was 3.3 mm (*CI: 1-5.5mm*) and the median visual acuity was 0.2 logMAR (*CI: 0-1.1*). Median dose at the apex of the tumor was 145 Gy (± *SD: 12.5 Gy*). The median scleral dose was 454.5 Gy (±*SD: 264.5 Gy*). The follow up examinations